MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 7650

The complete requirements for acquiring the electron tube described herein shall consist of this specification sheet and the latest issue of MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal

See figure 1
Mounting position: Any
Weight: 12 ounces nominal

ABSOLUTE RATINGS:

Parameter: / Unit:	Ef V (Note 1)	Eb Vdc	Ec2 Vdc	Ec1 Vdc	Ib mAdc	Ic1 mAdc
Maximum:						
AF power amp & modulator:	6.3 ±10%	3,000	1,200	---	500	100
Linear rf power amp:	6.3 ±10%	2,500	1,200	---	Note 2	100
Anode mod rf power amp-class C telephony:	6.3 ±10%	2,000	1,200	-250	500	100
RF power amp & osc-class C telegraphy:	6.3 ±10%	2,500	1,200	-250	500	100
TEST CONDITIONS:	6.3	2,500	400	Vary	240	---

ABSOLUTE RATINGS:

Parameter: / Unit:	Pin W	Pg2 W	Pp W	tk sec	Rg1 Ohms	T(seal) °C	Barometric pressure, reduced mmHg
Maximum:							
AF power amp & modulator:	1,500	25	600	120	15 K	250	140.7
Linear rf power amp:	1,250	25	600	120	15 K	250	140.7
Anode mod rf power amp-class C telephony:	1,000	17	400	120	15 K	250	140.7
RF power amp & osc-class C telegraphy	1,250	25	700	120	15 K	250	140.7
TEST CONDITIONS:	---	---	---	180	---	250(max)	---

GENERAL:

Qualification - Required

(B) denotes changes

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1266	Total grid current	4	Temperature anode core 250°C (min) 275°C (max)	0.65	II	Ic1	---	-50	µAdc
1261	Electrode voltage (1) (grid)			0.65	II	Ec1	-20	-42	Vdc
1256	Electrode current (screen)			0.65	II	Ic2	-8	+5	mAdc
1261	Electrode voltage (2) (grid)		Ec1/Ib = 5 mAdc	0.65	II	Ec2	---	-87	Vdc
1231	Pulsing emission	3, 4	eb = ec2 = ec1 = 850 v	0.65	II	Is	80	---	a
1372	Current division (method A, long pulse)		Eb = Ec2 = 500 Vdc; Ec1 = -100 Vdc; egk/ib = 2.5 a; prr = 10 to 12; tp = 2,000 to 2,500 µs	0.65	II	egk Ic1 Ic2	0 --- ---	25 280 230	v ma ma
---	Zero bias anode current	11	Eb = 500 Vdc; Ec2 = 450 Vdc; Ec1 = -100 Vdc; prr = 10 to 12; tp = 2,000 to 2,500 µs	0.65	II	Ib	1.4	---	a
	Quality conformance inspection, part 2								
1266	Primary grid emission (control)		Ec1/Pg1 = 10 watts; t = 300; anode and screen grounded; T(seal) = 100 ±15°C	---	---	Isg1	---	-40	µAdc
1266	Primary grid emission (screen)		Ec2/Pg2 = 25 watts; t = 300; anode and grid grounded; T(seal) = 100 ±15°C	---	---	Isg2	---	-40	µAdc
1301	Heater current			---	---	If	6.9	8.3	A
1366	Interelement leakage resistance, cold	5	Supply voltage = 200 Vdc; Rs = vary to give full scale deflection with external leads shorted together	---	---	R	8.0	---	Meg

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 2 - Continued								
1331	Direct-interelectrode capacitance		Use shield adapter RCA Model No. 7506, or equivalent; all unused electrodes grounded	---	---	Cg1-k	26	32	pF
						Cg1-g2	34	41	pF
						Cg2-p	4.3	6.3	pF
						Cg1-p	---	0.12	pF
						Cp-k	---	0.011	pF
						Cg2-k	---	1.1	pF
---	Power output	10	Ef = 5.7 v; Ec1/Ib = 500 mAdc; F = 400 ±15 MHz; t = 120	---	---	Po	600	---	W (useful)

Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3						
---	Life-test provisions		Group D; power output (1); t = 500 hours	---	---	---	---
---	Life-test end point		Power output (1)	Po	500	---	W (useful)
1002	Barometric pressure, reduced	9	Pressure = 140.7 mmHg; 3,000 Vdc applied between a and k, with k, g1, and g2 tied together	---	---	---	---
1042	Shock (1) specified pulse	7, 9	Test condition A	---	See note 6		---
1042	Shock (2) specified pulse	9, 12	Test condition D	---	See note 6		---
(B) 1032	Vibration, mechanical	6, 8, 9	RL = 2,000 ohms	Ep	---	40	Vac

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NOTES:

1. Because the cathode is subjected to considerable back bombardment as the frequency is increased with resultant increase in temperature, the heater voltage should be reduced depending on operating conditions and frequency to prevent overheating the cathode and resultant short life.

2. The maximum dc anode current at peak of envelope is 700 mAdc for a signal having a minimum peak-to-average power ratio of 2. During short periods of circuit adjustment under "Single-Tone" conditions, the average anode current may be as high as 700 mAdc. The maximum rating for a signal having a minimum peak-to-average power ratio less than 2, such as is obtained in "Single-Tone" operation, is 500 mAdc.

3. Read peak value of cathode current pulse after it has stabilized. Time limit for test is 1 minute from the time high voltage is applied. The pulse duration measured at 5 percent of the amplitude shall be not less than 3 µs. At 50 percent amplitude the duration shall be less than 2 µs. The applied voltage shall have a maximum repetition rate such that the duty cycle, based on the pulse length measured at 50 percent amplitude, shall not exceed 0.0002 (0.02 percent).

4. The following tests shall be the first tests performed after the holding period in the following order: Total grid current, pulsing emission.

5. The tube shall be cooled (room ambient temperature) 30 minutes before performing this test. The resistance between any two electrodes (except across the heater terminals) shall be measured in both directions.

6. There shall be no shorts or opens after this test. Tubes shall meet the limits for the following tests after this test: Total grid current, electrode voltage (1) (grid), electrode voltage (2) (grid), and electrode current (screen).

7. The tube shall be subjected to 18 impact shocks (see figure 2 for orientation of axes) as follows: Three shocks in each of three mutually perpendicular axes in the following sequence: Y+, Y-, X+, X-, Z+, and Z-. This is not a destructive test.

8. Vibrate along each of the three mutually perpendicular axes (X, Y, and Z, see figure 2) as follows:

Frequency (Hz)	Double amplitude (inches)	Acceleration (G)	Approx sweep duration (minutes)
5-10	0.08 ±10%		1
10-15		0.41 ±10%	0.5
15-105	0.036 ±10%		2.5
105-2,000		20 ±10%	4

Duration of the cycle shall be at least 8 minutes from 5 to 2,000 Hz. The frequency shall vary approximately logarithmic. This is not a destructive test.

An oscilloscope having a vertical amplifier with a minimum bandpass of dc to 10 MHz, shall be used to monitor the signal across the 2K ohm series anode resistor. The horizontal scope sweep frequency shall be approximately 1 cm per second. (Full scale vertical deflection shall be at least 250 volts.) The signal shall be dc coupled to the oscilloscope. If a signal greater than 60 volts occurs during the vibration sweep, the tube shall be considered a failure for this test. Sixty volts on the scope approximates 40 volts rms on the Bruel and Kjaer recorder, or equivalent.

The 60 volts referred to herein is deflection only, and does not include the oscilloscope signal resulting from the 10 ma anode current flowing through Rp.

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NOTES: -Continued.

(B) 9. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted. Shock, barometric pressure, and vibration samples may be formulated by combining tube types of the same basic tube type 7650 or 7651 construction. Sample may be composed of either type or a combination of basic generic types.

10. Use cathode drive cavity. Driving power = 35 watts; vary Rk to obtain Ib = 500 mAdc. Vary the loading to obtain maximum power output. Power output is the power delivered by the cavity to a transmission line terminated by the load.

11. Drive grid positive to zero volts with a square pulse. Zero bias anode current is the peak anode current at the start of the pulse.

12. The tube shall be subjected to five blows of any sequence in each of these positions: Y^+, X^+, Z^+, and Z- (see figure 2). This is not a destructive test.

User activities:
Navy - AS, OS, MC, CG, SH

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-N177)

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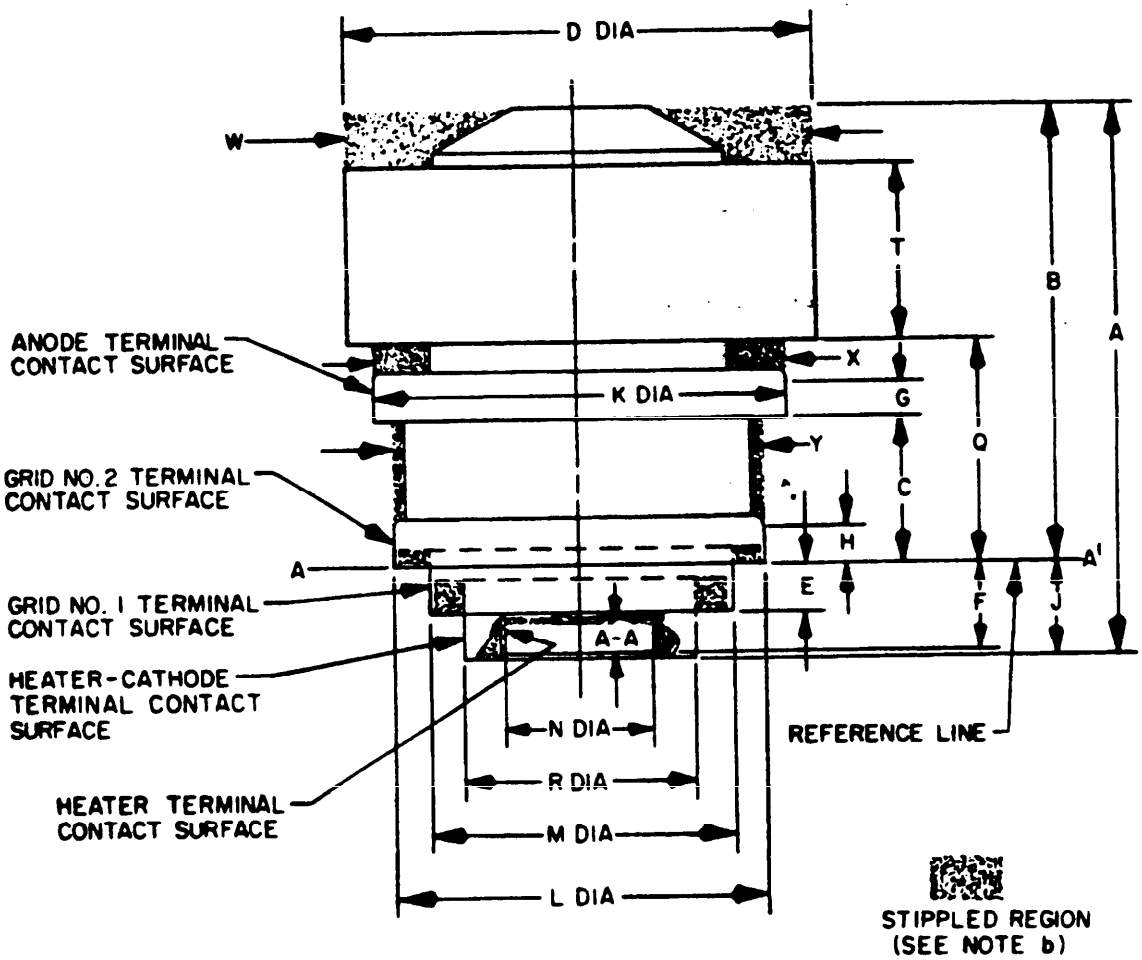


FIGURE 1. Outline drawing of electron tube type 7650.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A		2.40 (61.0)	
B	1.90 (48.3)	1.98 (50.3)	
C	.550 (13.98)	.600 (15.24)	
D	2.05 (52.1)	2.08 (52.8)	g
E	.18 (4.6)	.22 (5.6)	
J	.38 (9.7)	.42 (10.7)	
K	1.745 (44.32)		f
L	1.59 (40.4)		h
M	1.29 (32.8)		e
N		.67 (17.0)	
Q	.86 (21.8)	.93 (23.6)	
R	.99 (25.2)		c
Quality conformance inspection, part 3			k
F	.360 (9.14)	.410 (10.41)	
G	.145 (3.68)		
H	.15 (3.8)		
T	.735 (18.67)	.775 (19.69)	
AA	.115 (2.92)		
Reference dimensions			
W			d
X			a
Y			j

NOTES:
a. On any one tube this dimension shall never be greater than K.
b. Keep all stippled regions clear. Do not allow contacts or circuit components to protrude into these annular volumes.
c. Dimension R applies over length J minus E.
d. On any one tube this dimension shall never be greater than D.
e. Dimension M applies over length E only.
f. Dimension K applies over length G only.
g. Dimension D applies over length T only.
h. Dimension L applies over length H only.
j. On any one tube this dimension shall never be greater than L.
(B) k. These dimensional tests shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with a combined acceptance number of one. In the event of failure, the test shall be made as a part of Quality Conformance Inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 7650 - Continued.

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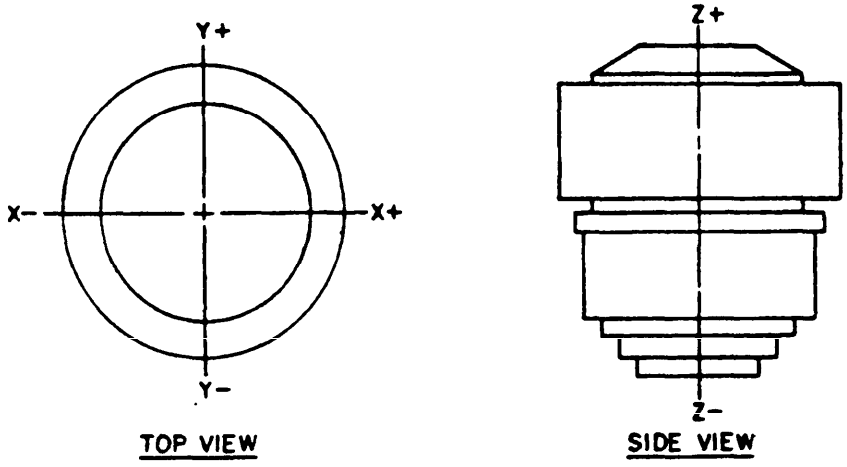
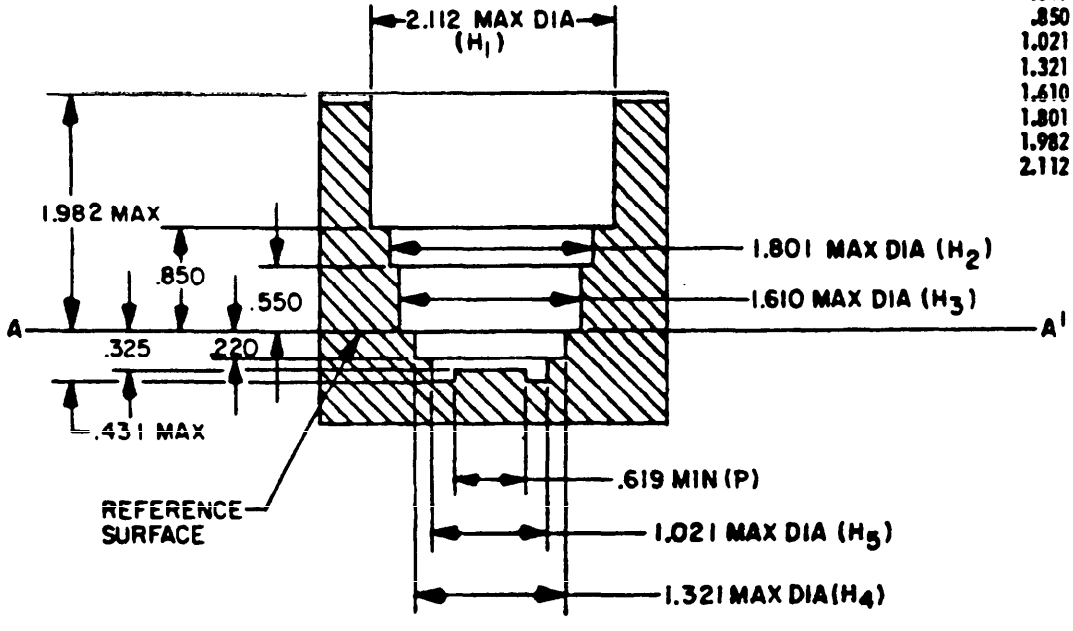


FIGURE 2. Orientation of axes.

INCHES	MM
.220	5.59
.325	8.26
.431	10.95
.550	13.97
.619	15.72
.850	21.59
1.021	25.93
1.321	33.55
1.610	40.89
1.801	45.75
1.982	50.34
2.112	53.64



NOTE:
With the cylindrical surfaces of the radiator band, anode terminal, grid No. 2 terminal, grid No. 1 terminal, heater-cathode terminal, and heater terminal clean, smooth and free of burrs, the tube will enter the gage, to determine out of roundness, concentricity and over size. Proper entry of the tube is obtained when the grid No. 2 terminal is seated on the shoulder marked A-A'. Seating is determined by failure of a .010" thickness gage 1/8" wide to enter more than 1/16" between shoulder surface and grid No. 2 terminal. Slots are provided to permit this measurement to be made.
The axes of cylindrical holes H_1 through H_5 and the axis of the post P shall be coincident within .001". Quality conformance inspection, part 1, AQL 1.0 percent, inspection level II.

FIGURE 3. Gage.

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